BioKEA BugPicker

Second Computer Setup Guide

> **Before starting:** confirm that commit `286f60c`, or a newer release, appears in the BioKEA/BugPicker GitHub repository. Never place a personal access token in this document or in the repository.

1. Prepare the OpenPnP scripts directory

```
mkdir -p ~/.openpnp2/scripts
cd ~/.openpnp2/scripts
ls -ld ~/.openpnp2/scripts/BugPicker
```

If BugPicker already exists, preserve it before cloning:

```
mv ~/.openpnp2/scripts/BugPicker ~/.openpnp2/scripts/BugPicker_before_github
```

2. Clone BugPicker

```
cd ~/.openpnp2/scripts
git clone https://github.com/BioKEA/BugPicker.git
cd BugPicker
```

If prompted, enter your GitHub username and use an approved personal access token as the password. Nothing is displayed while pasting a terminal password.

> Use `git clone`, not Download ZIP. Cloning preserves repository structure and supports later updates.

3. Verify the location

```
pwd
ls
```

The path should end in `/.openpnp2/scripts/BugPicker`. Confirm that `RUN_BUGPICKER.js`, numbered scripts, `requirements.txt`, and `Data/` are present.

4. Create the Python environment

```
cd ~/.openpnp2/scripts/BugPicker
python3 -m venv .venv
.venv/bin/pip install --upgrade pip
.venv/bin/pip install -r requirements.txt
```

If the venv module is unavailable:

```
sudo apt update
sudo apt install python3-venv
```

Then repeat the environment commands.

5. Verify active models

```
ls -lh \
  Data/insect_debris_classifier.pt \
  Data/taxonomy_classifier.pt \
  Data/qa_feedback/models/nozzle_qa_classifier.pt \
  Data/qa_feedback/models/well_qa_classifier.pt
```

All four files must be listed. If one is missing, confirm it was committed and pushed from the first computer.

6. Configure OpenPnP hardware

GitHub intentionally does not install machine-specific configuration or calibration. Configure the second system's cameras, USB devices, axes, soft limits, head, nozzle, vacuum actuator, trays, plates, and recovery plate.

> Do not blindly copy `machine.xml` when hardware ports, camera paths, or motion limits may differ. A matching machine configuration is only a starting point and must be verified before motion.

Verify the names expected by BugPicker:

- Head `H1`
- Pick nozzle `N1`
- Cameras `Top`, `Bottom`, and the configured HiRes camera
- BugPicker vacuum actuator

7. Calibrate the second machine

Use BugPicker setup/calibration to configure sorting-tray bounds and pick height, plate A1, well pitch, Top-camera plate A1, HiRes plate position and focus, bottom inspection, recovery plate, motion limits, and travel height.

This creates local files:

```
~/.openpnp2/scripts/BugPicker/multitray_calibration.json
~/.openpnp2/scripts/BugPicker/training_tray_calibration.json
```

These are ignored by Git because they are machine-specific.

8. Run software checks

```
cd ~/.openpnp2/scripts/BugPicker
python3 -m py_compile -- *.py
git status -sb
```

9. Test safely

1. Confirm Top, Bottom, and HiRes live feeds.
2. Verify calibration moves at travel height.
3. Run a dry or limited scan.
4. Test bottom-nozzle QA with empty and occupied states.
5. Test Lumen well detection and conditional HiRes imaging.
6. Complete one pick and placement before attempting a full plate.

> Keep the machine stop control immediately available during initial motion tests.

10. Install future updates

```
cd ~/.openpnp2/scripts/BugPicker
git status -sb
git pull --ff-only origin main
.venv/bin/pip install -r requirements.txt
```

Restart OpenPnP after updating. If Git reports local changes, review them; do not use `git reset --hard`.

Final Checklist

- ☐ Correct repository path
- ☐ Python environment created
- ☐ Requirements installed
- ☐ Four active models present
- ☐ Devices configured
- ☐ Local calibration completed
- ☐ Camera feeds and safe motion verified
- ☐ One-pick test completed